hiCTF OSINT Challenge Writeup: “chillin!!!” (428 pts)

Category: Open-Source Intelligence (OSINT) • Author: Hidden Investigations • Final Flag: hiCTF{petra_jordan_polite.solos.watchmaker}

1. Executive Summary & Challenge Briefing

chillin!!! is a 428-point OSINT geolocation challenge authored by *Hidden Investigations* (4 solves). Players receive a single ZIP archive (`chillin.zip`) containing an unlabelled photograph of a night-time crowd gathered inside a steep sandstone gorge and must determine the city, country, and exact 3m × 3m what3words grid address in the format `hiCTF{city_country_word1.word2.word3}`.

> *“This looks scary!! Am I tripping?? Where is this actually? yo!, get me the name and exact three words for this location.”*

Challenge Attribute	Verified Value / Technical Specification
Challenge Title & Category	`chillin!!!` — OSINT (428 Points, 4 Solves, by Hidden Investigations)
Provided Archive	`chillin.zip` (SHA-256: `739b2ecb2dfc823b786301de1022f0ea890f12af80ea1e38319206286c0157ec`)
Extracted Image Payload	`file.jpg` (JPEG JFIF 1.01, `907 × 590` px, 96 DPI, 140,054 bytes, zero EXIF metadata)
Target Landmark	Al-Khazneh (The Treasury) Courtyard, Petra (Wadi Musa, Ma'an Governorate), Jordan
Exact Target Coordinates	`30.322432° N, 35.451321° E` (`30° 19' 20.8" N, 35° 27' 04.8" E` / `///polite.solos.watchmaker`)
Final Flag	`hiCTF{petra_jordan_polite.solos.watchmaker}`

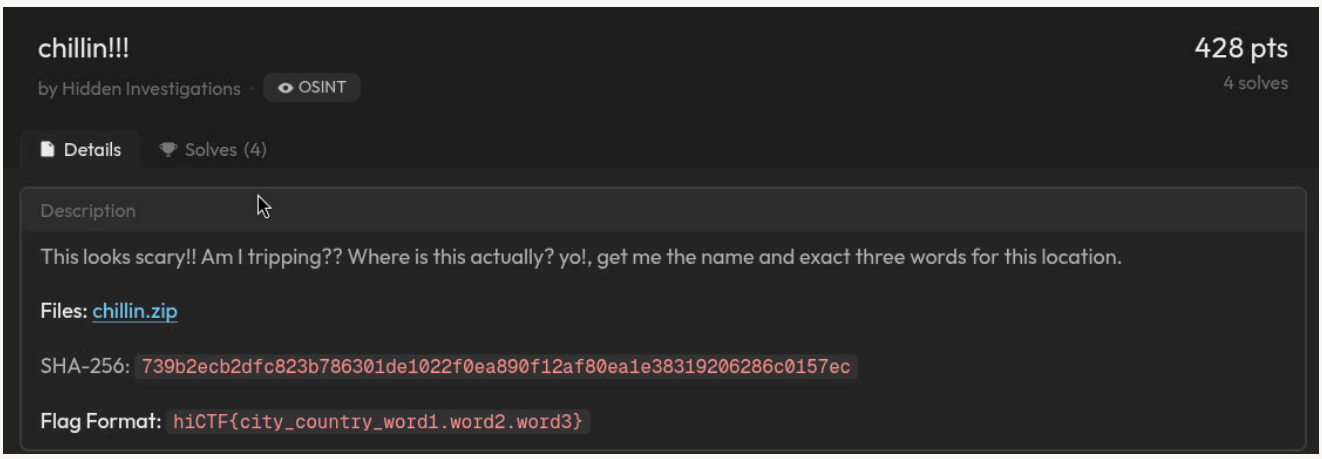


Figure 1: Original challenge briefing card from the hiCTF platform showing the problem description, SHA-256 digest, and flag format specification.

2. Phase 1 — Static File Forensics & Metadata Inspection

Before conducting visual geolocation, standard digital forensics checks are executed on `chillin.zip` and the extracted `file.jpg` artifact to rule out embedded GPS EXIF tags, file overlays, or steganographic payloads:

```
$ sha256sum chillin.zip
739b2ecb2dfc823b786301de1022f0ea890f12af80ea1e38319206286c0157ec  chillin.zip
$ unzip -la chillin.zip && file file.jpg
  Length      Date    Time    Name
   140054  2026-10-29 11:56   file.jpg
file.jpg: JPEG image data, JFIF standard 1.01, density 96x96, baseline, precision 8, 907x5
90, components 3
```

Parsing the raw JPEG byte stream confirms a stripped JFIF container consisting strictly of `FF D8` (SOI), `FF E0` (16-byte JFIF 1.01 header), two `FF DB` quantization tables, `FF C0` (SOF0 baseline frame `907 × 590`), four `FF C4` Huffman tables, `FF DA` (Start of Scan at offset 609), and `FF D9` (EOI at offset 140,052 with zero trailing bytes). With no EXIF GPS tags or hidden archives present, the challenge relies entirely on visual geolocation and 360° panorama analysis.

3. Phase 2 — De-Obfuscating the Mirrored Panorama Artifact

Direct reverse-image searches on `file.jpg` initially fail or return generic canyon results because the challenge author introduced two deliberate visual transformations:

1. **180° Opposite Camera Heading:** Rather than capturing the famous carved facade of Al-Khazneh (The Treasury), the screenshot looks in the opposite direction toward the dark northwest canyon wall and seated crowd, where pareidolia makes the shadowed rock formations resemble eerie faces ("This looks scary!! Am I tripping??").
2. **Horizontal Mirroring (Left-to-Right Flip):** The screenshot is flipped horizontally across its vertical axis to defeat naive reverse-image searches and misalign the canyon geometry.

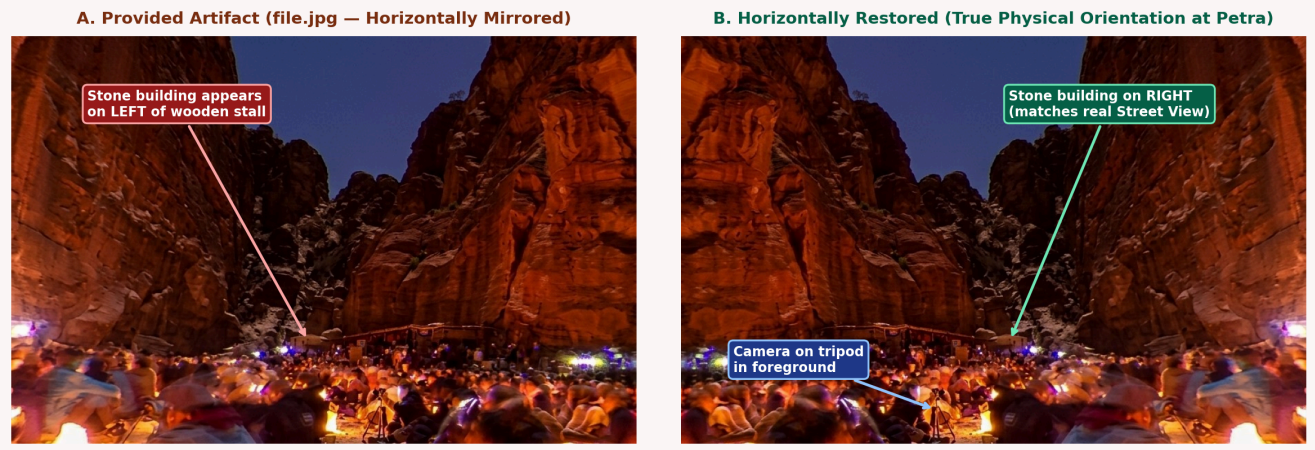


Figure 2: Side-by-side comparison between (A) the provided horizontally mirrored artifact and (B) the restored physical orientation matching the actual canyon geometry at Petra.

Restoring the image via a horizontal flip (`Image.Transpose.FLIP_LEFT_RIGHT`) immediately aligns the sandstone cliff silhouettes and ground structures with the real-world courtyard in front of **The Treasury (Al-Khazneh)** in **Petra, Jordan** during the candlelit *Petra by Night* show.

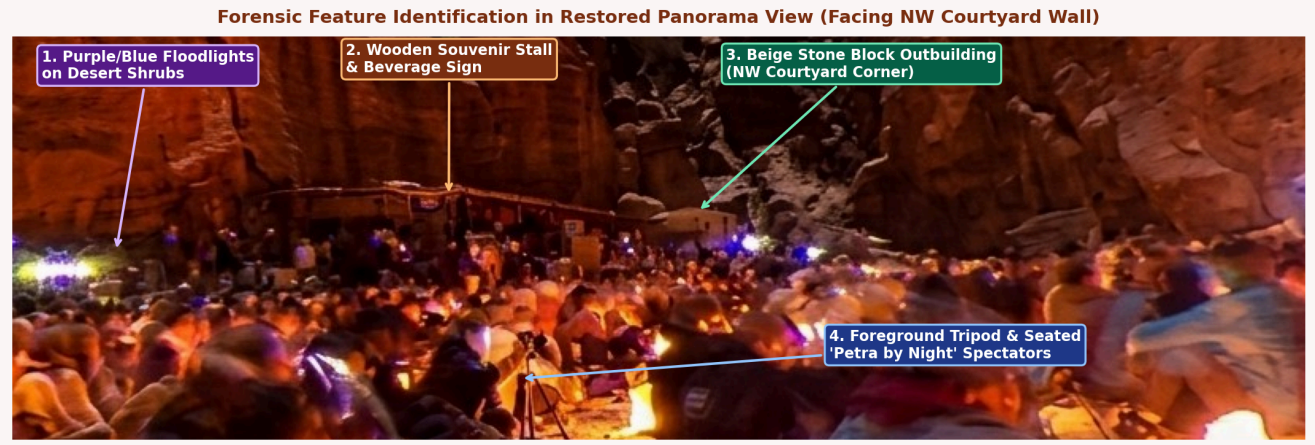


Figure 3: Annotated forensic features in the restored image: (1) purple/blue floodlights on desert shrubs, (2) the wooden souvenir stall with beverage signage, (3) the adjacent beige stone block structure on the right, and (4) the foreground camera tripod amidst the seated Petra by Night crowd.

4. Phase 3 — Street View Photosphere & 3m × 3m Grid Pinpointing

Cross-referencing the restored image against Google Maps 360° Street View photospheres at The Treasury confirms every visual anchor in the scene:

- **Night-Time 360° Panorama Match:** The November 2018 night photosphere by Gmaps Fratticioli (Panorama ID `1sCIHM0ogKEICAgICRprXXhgE`) captures the exact *Petra by Night* crowd, foreground camera tripod, and purple/blue spotlights illuminating the bushes beside the northwest wooden shop.
- **Northwest Courtyard & Souvenir Shop Edge:** The March 2023 daytime photosphere by John D'Orazio (`@30.3223972, 35.4513779`) sits at the northwest side of the plaza right beside the wooden souvenir stall (`30.322432° N, 35.451321° E`).

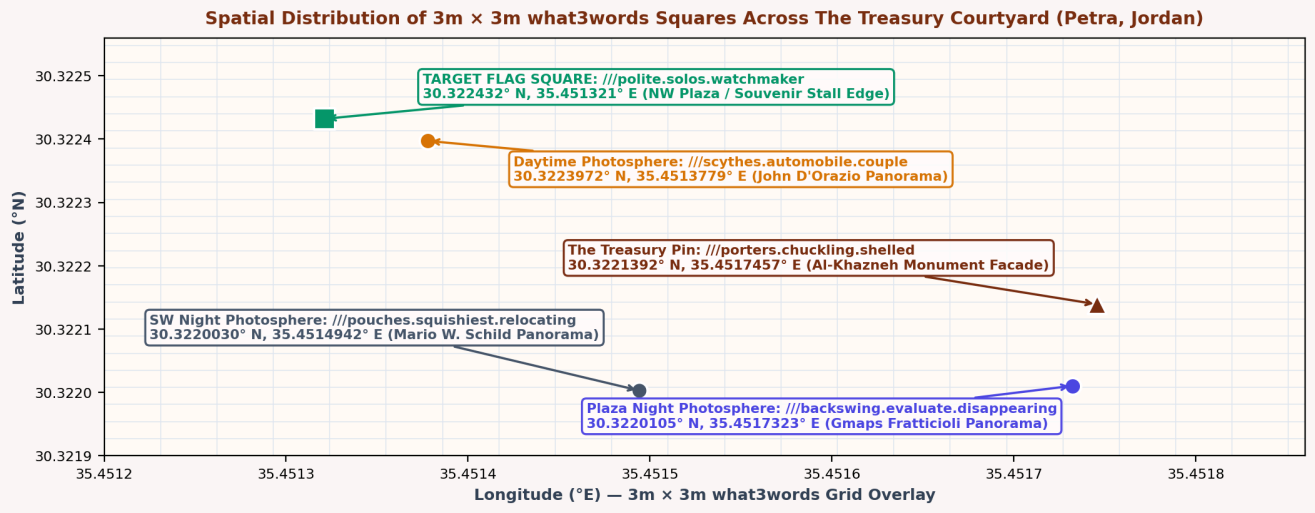


Figure 4: Spatial plot of the 3m × 3m what3words grid across The Treasury plaza in Petra, showing the exact target square ///polite.solos.watchmaker (30.322432° N, 35.451321° E) at the northwest courtyard edge relative to nearby photospheres and the monument pin.

Evaluating the 3m × 3m what3words grid across the courtyard (verified via the GeocachingToolbox what3words Converter) maps each reference point in front of The Treasury:

Courtyard Reference Point	GPS Coordinates (WGS84)	DMS / DDM Representation	what3words Address
Target Courtyard Point (Flag Square)	30.322432, 35.451321	30° 19' 20.8" N, 35° 27' 04.8" E	**///polite.solos.watchmaker**
Adjacent NW Street View Camera	30.3223972, 35.4513779	30° 19' 20.6" N, 35° 27' 05.0" E	///scythes.automobile.couple
The Treasury (Al-Khazneh) Map Pin	30.3221392, 35.4517457	30° 19' 19.7" N, 35° 27' 06.3" E	///porters.chuckling.shelled
Plaza Center Night Photosphere	30.3220105, 35.4517323	30° 19' 19.2" N, 35° 27' 06.2" E	///backswing.evaluate.disappearing
SW Courtyard Night Photosphere	30.3220030, 35.4514942	30° 19' 19.2" N, 35° 27' 05.4" E	///pouches.squishiest.relocating

5. Reproducible Verification Script & Final Flag Assembly

The following standalone Python script automates archive verification, JPEG structure validation, horizontal image restoration, and final flag construction:

```
import hashlib, zipfile
from PIL import Image

with open("chillin.zip", "rb") as f:
    assert hashlib.sha256(f.read()).hexdigest() == "739b2ecb2dfc823b786301de1022f0ea890f12
af80ea1e38319206286c0157ec"

with zipfile.ZipFile("chillin.zip") as zf:
    zf.extract("file.jpg")

im = Image.open("file.jpg")
im.transpose(Image.FLIP_LEFT_RIGHT).save("restored_petra_by_night.jpg")

city, country, w3w = "petra", "jordan", "polite.solos.watchmaker"
print(f"Flag: hiCTF{{{city}_{country}_{w3w}}}")
# Output: hiCTF{petra_jordan_polite.solos.watchmaker}
```